

Audio Amplifier Design and Implementation

Chandini Gayathri Jakku

Roll No: 2024102020

Dept. of ECE
IIIT Hyderabad

Veekshita Sai Viswanadham

Roll No: 2024102026

Dept. of ECE
IIIT Hyderabad

Abstract—Audio signals obtained from microphones or other input sources usually have very small amplitudes and cannot directly drive loads such as speakers. Therefore, amplification is required to increase the signal strength and deliver sufficient power while preserving the quality of the signal.

This project presents the design, simulation, and hardware implementation of a multi-stage audio amplifier built to meet specific electrical requirements. The amplifier consists of four main stages: a differential pre-amplifier, a common-emitter gain stage, an active band-pass filter, and a Class AB power amplifier. The pre-amplifier provides the initial amplification while improving noise immunity for weak input signals. The common-emitter stage further increases the voltage gain so that the overall gain requirement is achieved. An active band-pass filter is used to limit the signal to the audible frequency range of 20 Hz–20 kHz, thereby reducing unwanted frequency components and distortion. Finally, a Class AB power amplifier stage provides sufficient output power to drive a $10\ \Omega$ load without introducing additional voltage gain.

The complete circuit is first analysed and verified using LTSpice simulations and then implemented in hardware. The performance of the amplifier is evaluated using parameters such as voltage gain, bandwidth, total harmonic distortion (THD), and slew rate to verify that the design satisfies the required specifications.

I. INTRODUCTION

Audio amplifiers are an essential part of many electronic systems that process sound signals. The electrical signals produced by microphones or other audio sources are usually very small, typically in the range of a few millivolts peak-to-peak. Such weak signals cannot directly drive output devices like speakers because they do not have sufficient voltage or power. Therefore, amplification is required to increase the strength of the signal while maintaining the original waveform and audio quality.

The goal of an audio amplifier is not only to increase signal amplitude but also to preserve signal fidelity, maintain a stable frequency response over the audible range, and deliver sufficient power to the output load. Achieving all these requirements using a single amplifier stage is difficult. A single stage may provide high gain but poor noise performance, or it may deliver power but lack sufficient amplification. Combining all these functions in one stage can also reduce stability and increase distortion.

For this reason, practical audio amplifiers are usually designed using multiple stages, where each stage performs a specific function. This approach allows the circuit to achieve

high gain, good noise performance, controlled bandwidth, and sufficient output power while maintaining stability.

A. Overall Amplifier Architecture

In this project, a four-stage amplifier architecture is used, where each stage is responsible for a particular task in the signal amplification process.

- **Pre-Amplifier (Differential Stage):** The pre-amplifier accepts the weak input signal and provides the initial voltage amplification. A differential configuration is used to improve noise rejection and maintain high input impedance, preventing loading of the input source.
- **Gain Stage (Common-Emitter Amplifier):** This stage provides the majority of the voltage amplification required in the system. Together with the pre-amplifier, the combined voltage gain is designed to achieve an overall gain of approximately 500. To avoid distortion and clipping, the total gain is distributed between the differential pre-amplifier and the common-emitter gain stage rather than attempting to achieve very high gain within a single stage.
- **Active Band-Pass Filter:** The amplified signal is passed through an active band-pass filter that limits the signal to the audible frequency range of 20 Hz to 20 kHz. This helps remove unwanted frequency components and improves overall signal quality.
- **Class AB Power Amplifier:** The final stage increases the output power of the signal so that it can drive a load of $10\ \Omega$. A Class AB configuration is used to reduce crossover distortion while maintaining better efficiency compared to Class A amplifiers.

Another important design consideration is impedance matching between stages. The output impedance of each stage should be significantly lower than the input impedance of the following stage to prevent signal loss due to loading.

The amplifier is first analysed and verified using LTSpice simulations. After confirming the expected behaviour in simulation, the circuit is implemented in hardware and its performance is evaluated using parameters such as gain, frequency response, distortion, and output power.

II. DESIGN SPECIFICATIONS

The audio amplifier was designed according to the specifications provided in the project description. These constraints

TABLE I
DESIGN SPECIFICATIONS

Parameter	Symbol / Condition	Value
Supply voltage	V_{CC}, V_{EE}	± 5 V
Input signal (peak-to-peak)	$V_{in,pp}$	10–20 mV
Total voltage gain	$G_1 \times G_2$	≥ 500
Frequency response	Pass-band	20 Hz–20 kHz
Output power	P_{out}	≥ 1.5 W
Load resistance	R_L	10 Ω
Filter in-band attenuation	—	None (0 dB)
Power amplifier voltage gain	—	≈ 1 (Unity)

define the operating conditions of the circuit and guide the design of each stage of the amplifier.

These specifications determine the requirements for each stage of the amplifier. The pre-amplifier and gain stage together must provide a large voltage gain so that the weak input signal can be amplified sufficiently. The filter stage ensures that only frequencies within the audible range are passed through the system. Finally, the power amplifier stage is designed to deliver enough output power to drive the load without introducing additional voltage gain.

It is important to note that achieving a gain of 500 with a 20 mV peak-to-peak input would ideally require an output swing close to the supply rails (approximately ± 5 V). In practice, transistors require a finite voltage margin to remain in the forward active region, and therefore the output cannot reach the supply rails exactly. As a result, the design aims to achieve the highest possible gain while ensuring that the transistors remain in their proper operating region without causing clipping of the output signal.

III. PRE-AMPLIFIER STAGE (DIFFERENTIAL AMPLIFIER)

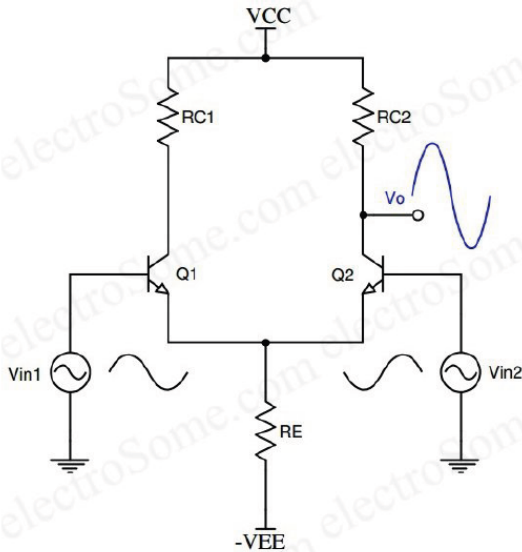


Fig. 1. Differential Pre-Amplifier

The pre-amplifier stage performs two important functions in the overall audio amplifier system: it provides the initial

voltage amplification of the weak input signal and suppresses noise through differential operation.

Audio signals obtained from microphones typically have very small amplitudes (in the millivolt range) and are highly susceptible to external noise. If the first stage of the amplifier introduces additional noise or loads the signal source heavily, the signal quality may degrade significantly. Therefore, the first stage must provide moderate gain while maintaining high input impedance and good noise rejection.

For this reason, a differential amplifier configuration is used. A differential amplifier amplifies the difference between two input signals while rejecting signals that appear identically at both inputs (common-mode signals). This property is quantified using the Common Mode Rejection Ratio (CMRR).

A. Circuit Description

The differential amplifier consists of two identical NPN transistors whose emitters are connected together through a common emitter resistance R_E . The collectors are connected to the positive supply through collector resistances R_{C1} and R_{C2} .

The output voltage is taken from the collector of one transistor. When the two transistors conduct equal currents, the circuit operates symmetrically and provides stable differential amplification.

B. Small Signal Analysis of Differential Stage

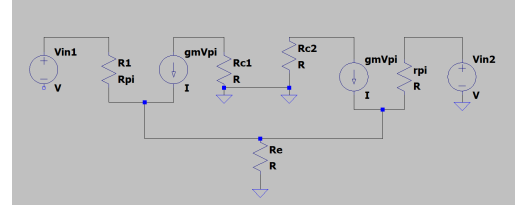


Fig. 2. Small signal equivalent circuit of the differential amplifier stage

To analyze the behaviour of the differential pre-amplifier, the hybrid- π small signal model of the transistors is used. In this model each transistor is represented by a base-emitter resistance r_π and a dependent current source $g_m v_{be}$.

For the differential pair, the emitters are connected through a common emitter resistance R_E . The small-signal currents through the transistors depend on the base-emitter voltages v_{be1} and v_{be2} .

Applying Kirchhoff's current law at the emitter node,

$$\frac{v_{be1}}{r_\pi} + g_m v_{be1} + \frac{v_{be2}}{r_\pi} + g_m v_{be2} = \frac{v_e}{R_E}$$

Factoring the terms,

$$(v_{be1} + v_{be2}) \left(\frac{1}{r_\pi} + g_m \right) = \frac{v_e}{R_E}$$

Since

$$v_{be1} = v_{in1} - v_e$$

$$v_{be2} = v_{in2} - v_e$$

Substituting,

$$(v_{in1} + v_{in2} - 2v_e) \left(\frac{1}{r_\pi} + g_m \right) = \frac{v_e}{R_E}$$

Solving for the emitter voltage,

$$v_e = \frac{(v_{in1} + v_{in2}) \left(\frac{1}{r_\pi} + g_m \right)}{\frac{1}{R_E} + 2 \left(\frac{1}{r_\pi} + g_m \right)}$$

For practical transistor operation

$$g_m r_\pi = \beta \gg 1$$

Thus

$$\left(\frac{1}{r_\pi} + g_m \right) \approx g_m$$

which simplifies the emitter voltage to

$$v_e \approx \frac{v_{in1} + v_{in2}}{2}$$

The output voltage is taken from the collector of the second transistor

$$v_{out} = -g_m(v_{in2} - v_e)R_C$$

Substituting the value of v_e ,

$$v_{out} = -g_m \left(v_{in2} - \frac{v_{in1} + v_{in2}}{2} \right) R_C$$

$$v_{out} = -\frac{g_m R_C}{2} (v_{in2} - v_{in1})$$

Thus the differential voltage gain becomes

$$A_v = \frac{v_{out}}{v_{in2} - v_{in1}} = -\frac{g_m R_C}{2}$$

This shows that the differential pair provides voltage gain proportional to the transconductance of the transistor and the collector resistance.

C. Input and Output Impedance

Since one input of the differential pair is driven while the other is grounded, the emitter node experiences a small-signal voltage approximately equal to half the input signal.

$$v_e \approx \frac{v_{in}}{2}$$

Thus the base-emitter voltage becomes

$$v_{be} = v_{in} - v_e = \frac{v_{in}}{2}$$

The input current is therefore

$$i_{in} = \frac{v_{be}}{r_\pi} = \frac{v_{in}}{2r_\pi}$$

Hence the effective input resistance is

$$R_{in} = r_\pi + \frac{r_\pi}{2}$$

$$R_{in} = \frac{3}{2}r_\pi$$

The output resistance is approximately equal to the collector resistance

$$R_{out} \approx R_C$$

which allows the amplified signal to be transferred to the next stage with minimal attenuation.

D. Biasing Requirements

For proper operation, both transistors must remain in the forward active region. This requires

$$V_{BE} > 0$$

$$V_{CE} > V_{BE}$$

Ensuring these conditions guarantees that the transistor operates as a controlled current source and provides linear amplification.

E. DC Analysis

Applying KVL in the emitter loop of the differential pair,

$$-0.7 - I_E R_E = -V_{EE}$$

Thus the emitter current is

$$I_E = \frac{V_{EE} - 0.7}{R_E}$$

Since the current splits equally between the two branches,

$$I_{E1} \approx I_{C1}, \quad I_{E2} \approx I_{C2}$$

$$I_C = \frac{I_E}{2}$$

Therefore,

$$I_C = \frac{V_{EE} - 0.7}{2R_E}$$

Substituting $V_{EE} = 5V$

$$I_C = \frac{5 - 0.7}{2R_E}$$

$$I_C = \frac{4.3}{2R_E}$$

$$I_C = \frac{2.15}{R_E}$$

F. AC Analysis

The transconductance of the transistor is

$$g_m = \frac{I_C}{V_T}$$

and the voltage gain of the differential amplifier stage is

$$A_v = g_m R_C$$

Substituting g_m

$$A_v = \frac{I_C}{V_T} R_C$$

The desired gain for this stage is approximately

$$A_v \approx 20$$

Thus

$$\frac{I_C R_C}{V_T} = 20$$

Substituting $I_C = \frac{2.15}{R_E}$

$$\frac{2.15 R_C}{R_E V_T} = 20$$

Using $V_T = 26mV$

$$2.15 R_C = 20 \times 26 \times 10^{-3} R_E$$

$$2.15 R_C = 0.52 R_E$$

Thus

$$R_E = \frac{2.15 R_C}{0.52}$$

$$R_E \approx 4.13 R_C$$

G. Component Selection

Assuming

$$I_C = 1mA$$

$$A_v = \frac{I_C R_C}{V_T}$$

$$20 = \frac{1mA \times R_C}{26mV}$$

$$R_C = \frac{20 \times 26mV}{1mA}$$

$$R_C \approx 520\Omega$$

Using the relation

$$R_E \approx 4.13 R_C$$

$$R_E = 4.13 \times 520$$

$$R_E \approx 2.15k\Omega$$

Due to non-ideal effects not accounted for in the basic equations, R_C was slightly adjusted to 560Ω through iteration to obtain the required gain.

H. LTSpice Simulation Results

The first stage of the amplifier was simulated in LTSpice to verify the behaviour of the differential pre-amplifier.

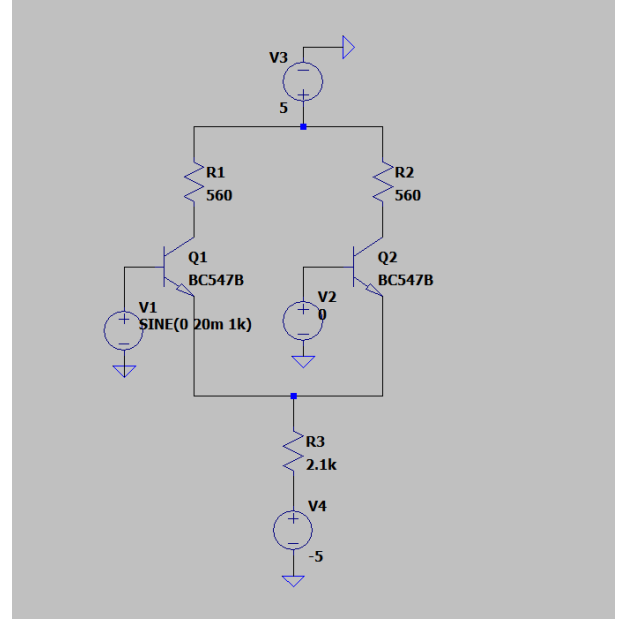


Fig. 3. LTSpice schematic of Stage 1 — Differential Pre-Amplifier

A sinusoidal signal of $10mV$ peak-to-peak at $1kHz$ was applied to the input of the circuit. The corresponding output waveform obtained from the simulation is shown in Figure 4.

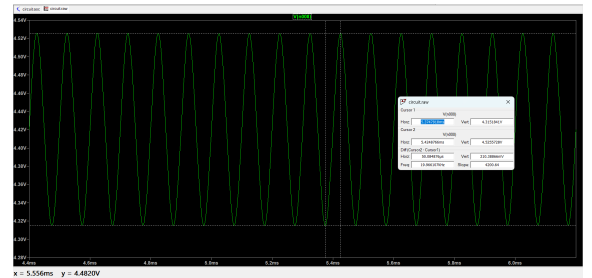


Fig. 4. Output waveform of Stage 1 (Differential Pre-Amplifier), Input: $10mV$, Frequency: $1kHz$

To verify the amplification behaviour, the input amplitude was increased to $20mV$ while maintaining the same frequency of $1kHz$. The resulting output waveform is shown in Figure 5.

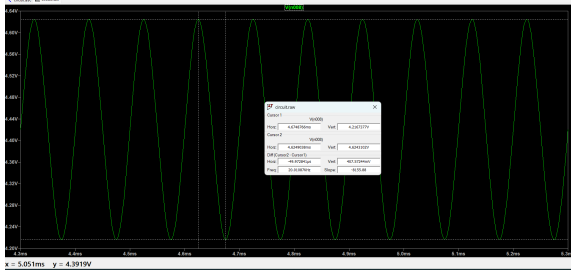


Fig. 5. Output waveform of Stage 1 (Differential Pre-Amplifier), Input: 20 mV, Frequency: 1 kHz

I. Hardware Measurements

After verifying the design through simulation, the circuit was constructed on hardware using discrete components. The input signal was generated using the waveform generator of the oscilloscope.

A sinusoidal signal of approximately 20 mV peak-to-peak at 1 kHz was applied to the circuit. The input and output signals were simultaneously observed on the oscilloscope.

The measured output signal had an amplitude of approximately 400 mV peak-to-peak, which corresponds to a voltage gain of

$$A_v = \frac{V_{out}}{V_{in}} \approx \frac{213 \text{ mV}}{10 \text{ mV}} \approx 21.3$$

This confirms that the hardware implementation achieves the desired amplification.

J. Common Mode Rejection

To evaluate the noise rejection capability of the differential amplifier, the common-mode gain was measured and compared with the differential gain.

The Common Mode Rejection Ratio (CMRR) is given by

$$CMRR = 20 \log \left(\frac{A_d}{A_c} \right)$$

where A_d is the differential gain and A_c is the common-mode gain.

Substituting the measured values,

$$CMRR = 20 \log \left(\frac{20}{0.1275} \right)$$

$$CMRR \approx 44 \text{ dB}$$

A high CMRR indicates that the amplifier effectively rejects common-mode noise, which is important when amplifying very weak input signals.

K. Simulation Results

The behaviour of the pre-amplifier stage was verified using hardware measurements with an oscilloscope.

A sinusoidal input signal of 10 mV peak-to-peak at 1 kHz was applied to the circuit.

Figure 6 shows the input waveform to the pre-amplifier, while Figure 7 shows the corresponding output waveform after amplification.

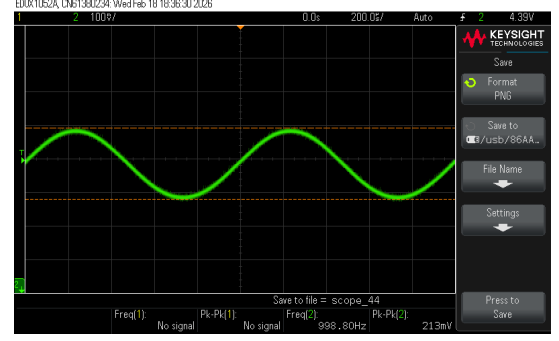


Fig. 6. output waveform to the pre-amplifier stage (input:10 mV, frequency:1 kHz)

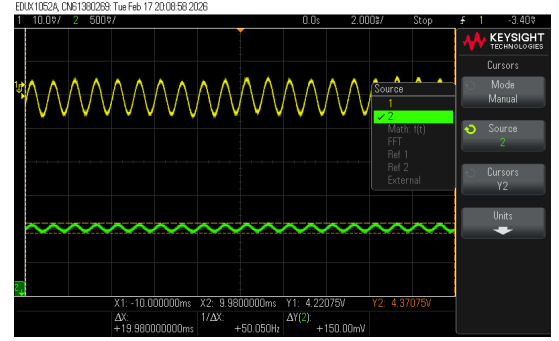


Fig. 7. Output waveform of the pre-amplifier stage (input:20 mV, frequency:1 kHz)

IV. GAIN STAGE (COMMON EMITTER AMPLIFIER)

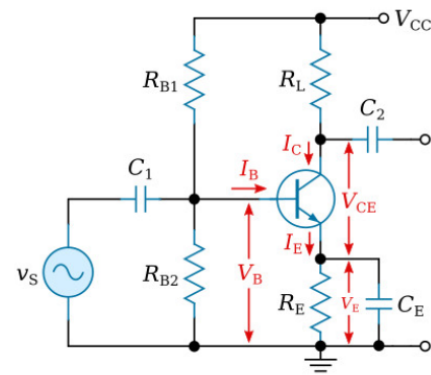


Fig. 8. Common Emitter Amplifier

After the initial amplification provided by the differential pre-amplifier, the signal is further amplified using a common-emitter (CE) amplifier. This stage is responsible for providing a large portion of the voltage gain required to achieve the overall gain specification of the amplifier.

The common-emitter configuration is widely used because it offers significant voltage gain with a relatively simple circuit

structure. A voltage divider bias network is used to ensure stable biasing so that the transistor operates in the forward active region during signal amplification.

A. Small Signal Model

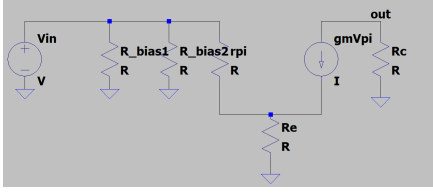


Fig. 9. Hybrid- π small signal model of the BJT gain stage

To analyse the behaviour of the gain stage, the hybrid- π small signal model of the BJT is used. In this model the transistor is represented by the base-emitter resistance r_π and a dependent current source $g_m v_\pi$.

The transconductance of the transistor is

$$g_m = \frac{I_C}{V_T}$$

where I_C is the collector current and V_T is the thermal voltage ($\approx 26mV$ at room temperature).

The base-emitter resistance is given by

$$r_\pi = \frac{\beta}{g_m}$$

where β is the current gain of the transistor.

B. Voltage Gain Derivation

In a common-emitter amplifier the small signal collector current is

$$i_c = g_m v_\pi$$

The output voltage is taken across the collector resistor R_C

$$v_o = -i_c R_C$$

Substituting for the collector current

$$v_o = -g_m v_\pi R_C$$

If an emitter resistance R_E is present and not completely bypassed, it reduces the effective gain. The voltage gain of the stage can therefore be written as

$$A_v = -\frac{R_C}{\frac{1}{g_m} + R_E}$$

The negative sign indicates that the output signal is inverted with respect to the input.

C. Input and Output Impedance

The input impedance of the gain stage must be sufficiently large so that it does not load the previous stage.

The input resistance is given by

$$R_{in} = R_1 || R_2 || r_\pi$$

The output impedance of the stage is approximately equal to the collector resistance

$$R_{out} \approx R_C$$

D. Biasing Design

The amplifier uses dual supply rails

$$V_{CC} = +5V$$

$$V_{EE} = -5V$$

A small quiescent collector current is chosen to minimise power dissipation

$$I_C \approx 0.2mA$$

Using the transconductance relation

$$g_m = \frac{I_C}{V_T}$$

$$g_m = \frac{0.2mA}{26mV}$$

$$g_m \approx 7.66mS$$

E. Collector Resistor Design

To allow maximum signal swing without clipping, the collector voltage is chosen near the midpoint of the supply.

$$V_C \approx 0.5V$$

The voltage drop across the collector resistor is therefore

$$V_{RC} = V_{CC} - V_C$$

$$V_{RC} = 5 - 0.5 = 4.5V$$

Using Ohm's law

$$R_C = \frac{V_{RC}}{I_C}$$

$$R_C = \frac{4.5}{0.2mA}$$

$$R_C \approx 22.5k\Omega$$

F. Base Biasing

The base voltage is selected as

$$V_B = -4V$$

Since the base-emitter junction drop is approximately

$$V_{BE} \approx 0.634V$$

the emitter voltage becomes

$$V_E = V_B - V_{BE}$$

$$V_E \approx -4.634V$$

Using the voltage divider equation

$$V_B = V_{EE} + \frac{R_2}{R_1 + R_2}(V_{CC} - V_{EE})$$

substituting the required base voltage gives

$$\frac{R_2}{R_1 + R_2} = 0.1$$

Thus the resistor ratio becomes

$$R_1 : R_2 = 9 : 1$$

Large resistor values are chosen to minimise the base current

$$R_1 = 180k\Omega$$

$$R_2 = 20k\Omega$$

G. Emitter Resistor Design

The emitter voltage relative to the negative rail is

$$V_E - V_{EE} = 0.366V$$

The emitter current is approximately

$$I_E \approx I_C = 0.2mA$$

Thus the total emitter resistance is

$$R_E = \frac{0.366}{0.2mA}$$

$$R_E \approx 1.83k\Omega$$

From the small signal analysis, the voltage gain of the common-emitter amplifier is

$$A_v = \frac{R_C}{\frac{1}{g_m} + R_{E1}}$$

Using the design values

$$R_C = 22.5k\Omega$$

$$g_m = 7.66mS$$

the intrinsic emitter resistance is

$$r_e = \frac{1}{g_m} \approx 130\Omega$$

The expected gain of this stage is approximately

$$A_v \approx 22.5$$

Substituting in the gain equation

$$22.5 = \frac{22500}{130 + R_{E1}}$$

Solving,

$$130 + R_{E1} \approx 1000$$

$$R_{E1} \approx 870\Omega$$

Thus the unbypassed emitter resistance is chosen as

$$R_{E1} = 870\Omega$$

Next, the total emitter resistance required for the DC bias condition is obtained from

$$V_E - V_{EE} = (R_{E1} + R_{E2})I_E$$

From the bias calculations

$$V_E - V_{EE} = 0.366V$$

$$I_E \approx 0.2mA$$

Thus

$$0.366 = (870 + R_{E2})(0.2mA)$$

Solving,

$$R_{E2} \approx 960\Omega$$

Therefore the emitter resistance is implemented as

$$R_{E1} = 870\Omega$$

$$R_{E2} = 960\Omega$$

where R_{E2} is bypassed by a capacitor so that it does not affect the AC gain of the amplifier. This allows R_{E1} to control the small-signal gain while R_{E2} helps establish the correct DC bias point.

Using the chosen resistor values

$$R_1 = 180k\Omega$$

$$R_2 = 20k\Omega$$

and

$$r_\pi \approx 38.4k\Omega$$

we obtain

$$R_{in} = 180k || 20k || 38.4k$$

$$R_{in} \approx 12k\Omega$$

This value is significantly larger than the collector resistance of the pre-amplifier stage, ensuring that the gain stage does not load the previous stage

H. Emitter Bypass Capacitor

A capacitor is connected in parallel with one of the emitter resistors so that the AC signal sees a much smaller impedance while the DC biasing remains unchanged.

A capacitor value of

$$C = 100\mu F$$

is chosen so that the reactance is very small within the audio frequency range. .

I. Practical Observations

During simulation it was observed that the output waveform initially exhibited clipping at the lower voltage levels. This indicated that the collector-base junction was becoming forward biased at extreme signal swings.

To address this issue, the emitter resistance used for quiescent biasing was slightly increased. After iterative adjustment, a value of

$$R_E = 1.1k\Omega$$

provided stable operation without clipping.

During simulation it was also observed that the required gain was not achieved with the initially calculated value of

$$R_C = 22.5k\Omega$$

After iterative adjustment in LTspice, a value of

$$R_C = 23k\Omega$$

was chosen as it provided the best gain without clipping.

With these values the measured voltage gain of the stage was

$$A_v \approx 22.95$$

which is close to the theoretical estimate of approximately 22.5. The small difference arises due to non-ideal and non-linear behaviour of the transistor.

J. LTSpice Simulation Results

The voltage gain obtained from LTSpice simulation is

$$A_v = \frac{9180mV}{20mV} \approx 459$$

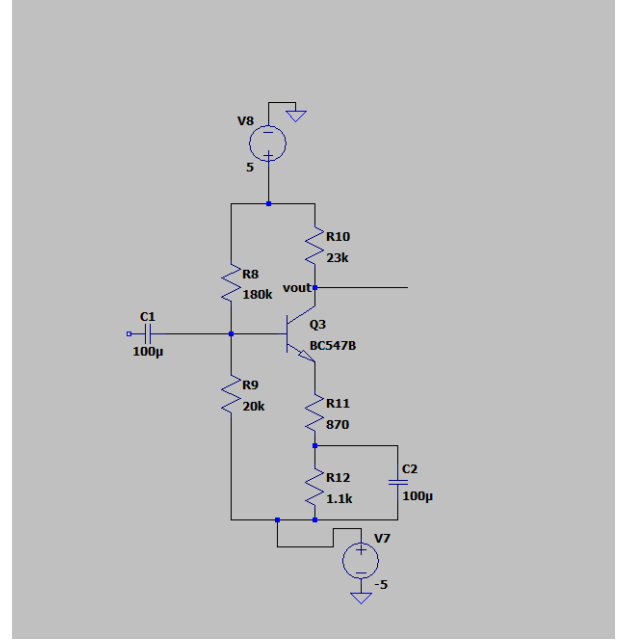


Fig. 10. LTSpice schematic of Stage 2 — Common-Emitter Gain Stage

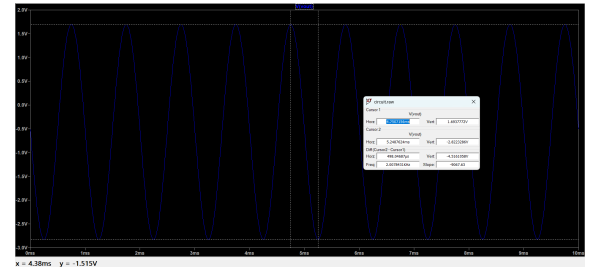


Fig. 11. Output waveform of Stage 2 (Input: 10 mV, Frequency: 1 kHz)

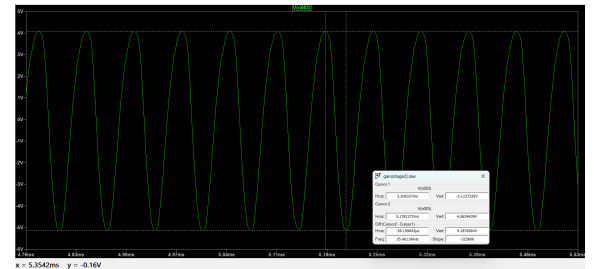


Fig. 12. Output waveform of Stage 2 (Input: 20 mV, Frequency: 1 kHz)

K. Hardware Simulation Results

The voltage gain measured in hardware is

$$A_v = \frac{8.8V}{20mV} \approx 440$$

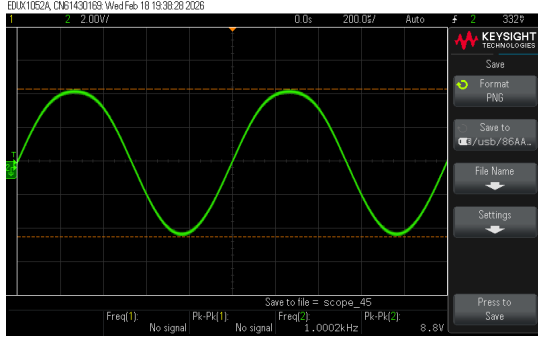


Fig. 13. Hardware output waveform of the gain stage (Input: 20 mV, Frequency: 1 kHz)

V. ACTIVE BAND-PASS FILTER

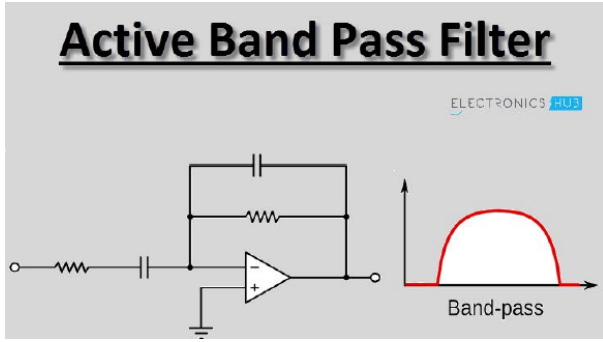


Fig. 14. Active band pass filter

After the gain stage, the amplified signal is passed through an active band-pass filter. The purpose of this stage is to limit the signal to the audible frequency range while attenuating unwanted low-frequency and high-frequency components.

The audible range of human hearing is approximately 20 Hz to 20 kHz. To ensure that the required signal band is passed without attenuation at the boundaries, the filter is designed with slightly wider cutoff frequencies

$$f_L = 10 \text{ Hz}$$

$$f_H = 40 \text{ kHz}$$

Thus the desired passband of 20 Hz–20 kHz lies comfortably within the filter response.

A. Circuit Analysis

The filter is implemented using an operational amplifier in an inverting configuration. The non-inverting terminal of the op-amp is connected to ground. Due to the virtual short property of the operational amplifier, the inverting terminal is also maintained at virtual ground.

Applying Kirchhoff's current law at the inverting node

$$V_{in} - 0 = IZ_1$$

$$0 - V_{out} = IZ_2$$

Thus the voltage gain becomes

$$\frac{V_{out}}{V_{in}} = -\frac{Z_2}{Z_1}$$

where

$$Z_1 = \frac{sR_1C_1 + 1}{sC_1}$$

$$Z_2 = \frac{R_2}{sR_2C_2 + 1}$$

Substituting these expressions gives the transfer function

$$H(s) = \frac{V_{out}}{V_{in}} = -\frac{sR_2C_1}{(1 + sR_1C_1)(1 + sR_2C_2)}$$

where $s = j\omega$.

B. Cutoff Frequency Relations

From the transfer function, the lower and upper cutoff frequencies are

$$f_L = \frac{1}{2\pi R_1 C_1}$$

$$f_H = \frac{1}{2\pi R_2 C_2}$$

C. Lower Cutoff Frequency Design

For the chosen lower cutoff frequency

$$f_L = 10 \text{ Hz}$$

$$R_1 C_1 = \frac{1}{2\pi f_L}$$

$$R_1 C_1 = \frac{1}{2\pi(10)}$$

$$R_1 C_1 \approx 2.6 \times 10^{-2}$$

Choosing

$$C_1 = 50 \text{ nF}$$

the resistor value becomes

$$R_1 = \frac{2.6 \times 10^{-2}}{50 \times 10^{-9}}$$

$$R_1 \approx 531 \text{ k}\Omega$$

D. Upper Cutoff Frequency Design

For the upper cutoff frequency

$$f_H = 40 \text{ kHz}$$

$$R_2 C_2 = \frac{1}{2\pi f_H}$$

$$R_2 C_2 = \frac{1}{2\pi(40 \times 10^3)}$$

$$R_2 C_2 \approx 5 \times 10^{-6}$$

For simplicity, the same resistor value is chosen

$$R_1 = R_2 = 531 \text{ k}\Omega$$

Thus

$$C_2 = \frac{5 \times 10^{-6}}{531 \times 10^3}$$

$$C_2 \approx 5 \text{ pF}$$

E. Key Parameters

Center Frequency

The center frequency of the band-pass filter is given by

$$f_0 = \frac{1}{2\pi\sqrt{R_1 R_2 C_1 C_2}}$$

This is the frequency at which the filter achieves maximum gain.

Bandwidth

The bandwidth of the filter is defined as

$$BW = f_H - f_L$$

which represents the range of frequencies that are passed by the filter.

Quality Factor

The quality factor of the filter is defined as

$$Q = \frac{f_0}{BW}$$

A larger value of Q corresponds to a narrower passband, while a smaller value indicates a wider frequency response.

F. Frequency Response

The magnitude of the transfer function can be written as

$$\left| \frac{V_{out}}{V_{in}} \right| = \frac{\omega R_2 C_1}{\sqrt{(1 + (\omega R_2 C_2)^2)(1 + (\omega R_1 C_1)^2)}}$$

The circuit behaves as a high-pass filter at low frequencies and as a low-pass filter at high frequencies, resulting in an overall band-pass response.

To obtain unity gain in the passband, the resistors are chosen such that

$$R_1 = R_2$$

Thus the midband gain becomes

$$A_v = 1$$

This ensures that the filter performs only frequency selection without introducing additional amplification.

G. LTSpice Simulation Results

The voltage gain obtained from LTSpice simulation is

$$A_v = 448$$

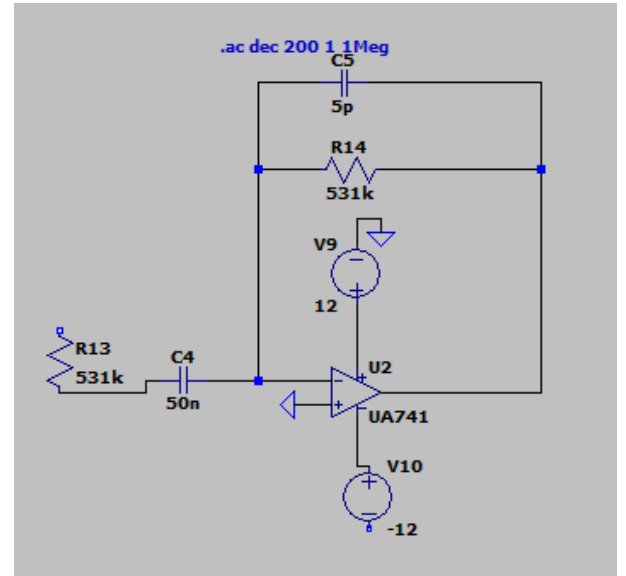


Fig. 15. LTSpice schematic of Stage 3 — Active Band-Pass Filter

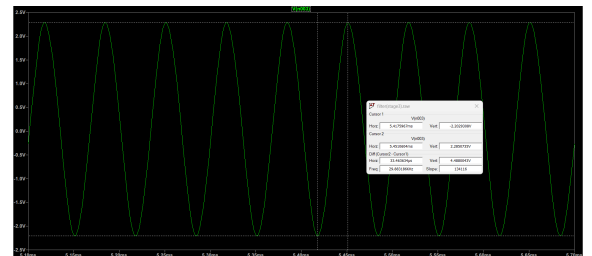


Fig. 16. Output waveform of Stage 3 (Input: 10 mV, Frequency: 1 kHz)

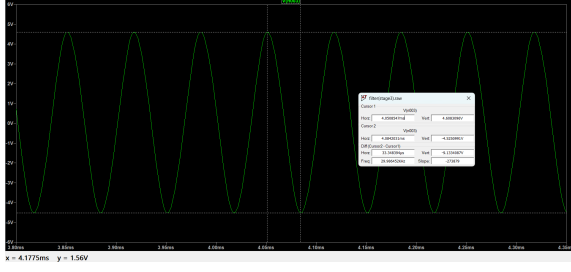


Fig. 17. Output waveform of Stage 3 (Input: 20 mV, Frequency: 1 kHz)

H. Hardware Simulation Results

The voltage gain measured in hardware is

$$A_v = 435$$

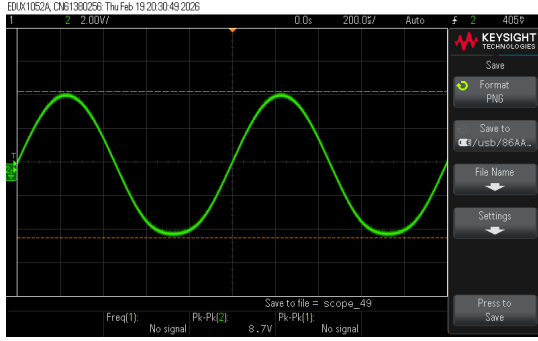


Fig. 18. Hardware output waveform of the active band-pass filter (Input: 20 mV, Frequency: 1 kHz)

VI. POWER AMPLIFIER

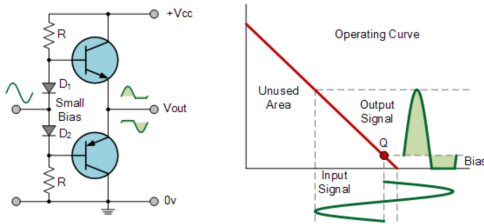


Fig. 19. Class AB Power Amplifier

A. Purpose of the Power Amplifier

The output of the previous stages of the amplifier possesses sufficient voltage amplitude but cannot directly drive the 10Ω load. The pre-amplifier, gain stage, and filter are primarily voltage amplifiers and therefore have relatively high output impedance. A low-resistance load such as a speaker requires significant current, which these stages cannot supply without severe signal attenuation.

For this reason, a dedicated **power amplifier stage** is required. The role of this stage is not to provide additional voltage gain, but rather to supply the current required by the load while maintaining signal fidelity.

B. Choice of Class AB Topology

The final stage uses a **Class AB complementary push-pull amplifier**. The selection of Class AB operation is motivated by the following considerations:

- **Class A:** The transistor conducts for the entire input cycle. Although distortion is minimal, the efficiency is very poor because large current flows even when no signal is present.
- **Class B:** Two complementary transistors conduct for opposite half cycles. This improves efficiency but causes *crossover distortion* near the zero crossing.
- **Class AB:** Both transistors are slightly biased into conduction so that each conducts for slightly more than half the cycle. This removes crossover distortion while keeping power dissipation moderate.

For this reason, a Class AB stage provides the best compromise between efficiency and signal quality for the final stage of the amplifier.

C. Circuit Topology

The circuit uses a complementary pair of power transistors:

- Q4 : TIP31A (NPN)
- Q5 : TIP32A (PNP)

Both transistors operate in a **common-collector (emitter follower)** configuration.

The important properties of an emitter follower are

$$A_v \approx 1$$

The output current is approximately

$$I_{out} \approx I_E$$

and

$$I_E \approx (\beta + 1)I_B$$

Thus the output current can be written as

$$I_{out} \approx (\beta + 1)I_B$$

Since the current gain of the transistor satisfies

$$\beta \gg 1$$

it is common to approximate

$$I_{out} \approx \beta I_B$$

Thus the stage provides large **current gain** while introducing almost no voltage gain. This satisfies the design specification that the power amplifier should not significantly increase the voltage gain of the system.

D. Bias Voltage using Diodes

To eliminate crossover distortion, the bases of the complementary transistors are biased using two silicon diodes. Each diode has an approximate forward voltage drop

$$V_D \approx 0.63V$$

Thus the total bias voltage across the diode string becomes

$$V_{bias} = 2V_D$$

$$V_{bias} = 2 \times 0.63$$

$$V_{bias} \approx 1.26V$$

This voltage is approximately equal to the sum of the base-emitter voltages of the two transistors

$$V_{BE1} + V_{BE2} \approx 1.2V$$

Therefore both transistors remain slightly forward biased even when the input signal is zero, eliminating crossover distortion.

E. Load Current Requirement

The amplifier must drive a load

$$R_L = 10\Omega$$

Assuming the output can swing approximately

$$V_{out,peak} \approx 5V$$

the peak load current is

$$I_{L,peak} = \frac{V_{out}}{R_L}$$

$$I_{L,peak} = \frac{5}{10}$$

$$I_{L,peak} = 0.5A$$

Thus the power transistors must supply peak currents of approximately

$$I_{L,peak} \approx 500mA$$

which justifies the use of power devices such as TIP31A and TIP32A.

F. Base Current Requirement

For an emitter follower, the output current is approximately

$$I_{out} \approx \beta I_B$$

From the datasheet a typical current gain is

$$\beta \approx 200$$

Thus the required base current is

$$I_B = \frac{I_C}{\beta}$$

$$I_B = \frac{0.5}{200}$$

$$I_B \approx 2.5mA$$

G. Bias Resistor Design

The diode bias network is supplied from $\pm 5V$ rails. Therefore the total voltage across the resistor-diode chain is

$$V_{total} = 10V$$

The two diodes drop approximately

$$2V_D = 1.26V$$

Thus the remaining voltage across the two resistors is

$$V_R = 10 - 1.26$$

$$V_R = 8.74V$$

Since the circuit is symmetric

$$V_{R15} = V_{R16} = \frac{8.74}{2}$$

$$V_{R15} = V_{R16} = 4.37V$$

To maintain stable biasing, the diode current should be several times larger than the base current.

Choosing

$$I_{bias} \approx 4.3mA$$

Using Ohm's law

$$R = \frac{V}{I}$$

$$R = \frac{4.37}{0.0043}$$

$$R \approx 1k\Omega$$

Thus the resistor values are chosen as

$$R15 = R16 = 1k\Omega$$

H. Input Coupling Capacitor

The input coupling capacitor prevents any DC component from the previous stage from affecting the bias of the power amplifier.

The cutoff frequency due to the coupling capacitor is

$$f_c = \frac{1}{2\pi RC}$$

Assuming the input resistance of the stage is approximately

$$R \approx 1k\Omega$$

and choosing

$$C_8 = 10\mu F$$

the cutoff frequency becomes

$$f_c = \frac{1}{2\pi(1000)(10 \times 10^{-6})}$$

$$f_c \approx 15.9Hz$$

Thus the capacitor allows the entire audio band (20Hz–20kHz) to pass without attenuation.

I. LTSpice Simulation Results

The voltage gain obtained from LTSpice simulation is

$$A_v = 422$$

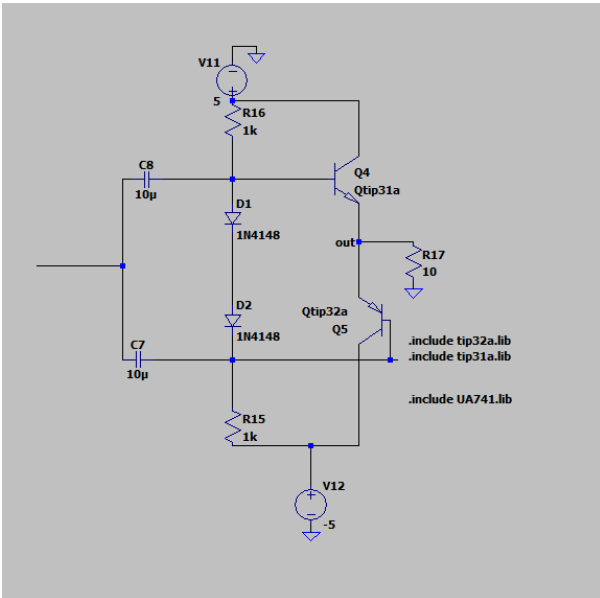


Fig. 20. LTSpice schematic of Stage 4 — Class AB Power Amplifier

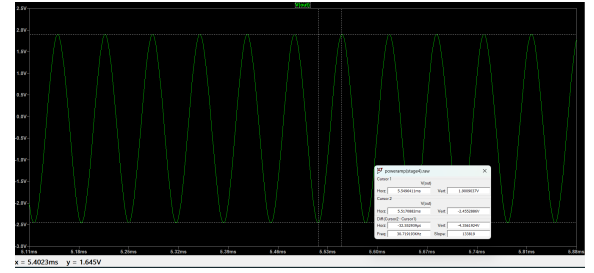


Fig. 21. Output waveform of Stage 4 (Input: 10 mV, Frequency: 1 kHz)

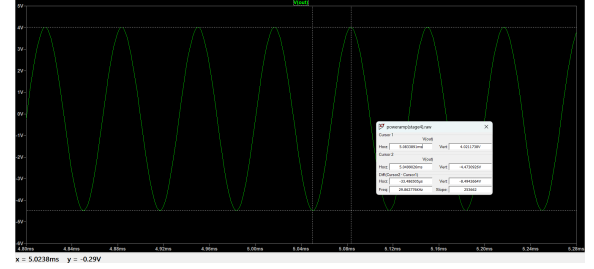


Fig. 22. Output waveform of Stage 4 (Input: 20 mV, Frequency: 1 kHz)

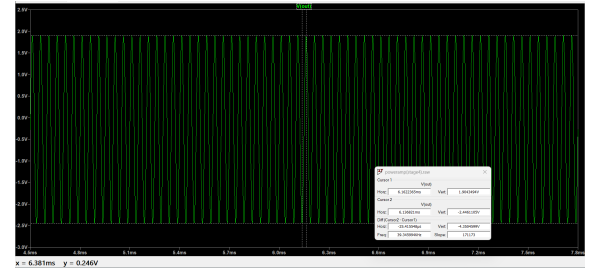


Fig. 23. Output waveform of Stage 4 (Input: 10 mV, Frequency: 10 kHz)

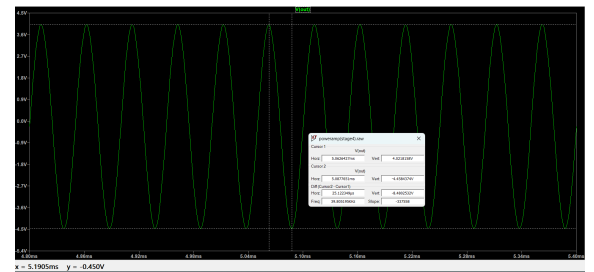


Fig. 24. Output waveform of Stage 4 (Input: 20 mV, Frequency: 20 kHz)

J. Hardware Simulation Results

The voltage gain measured in hardware is

$$A_v = 410$$

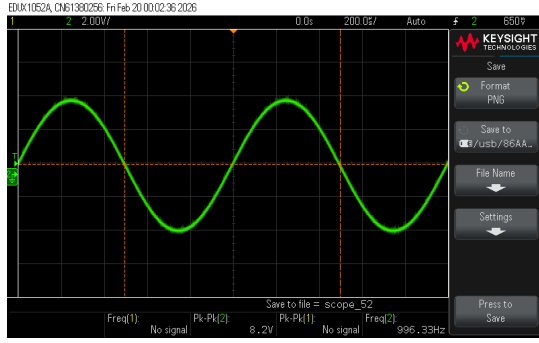


Fig. 25. Hardware output waveform of the power amplifier (Input: 20 mV, Frequency: 1 kHz)

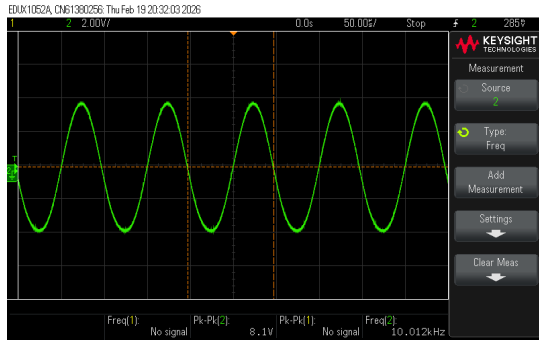


Fig. 26. Hardware output waveform of the power amplifier (Input: 20 mV, Frequency: 10 kHz)

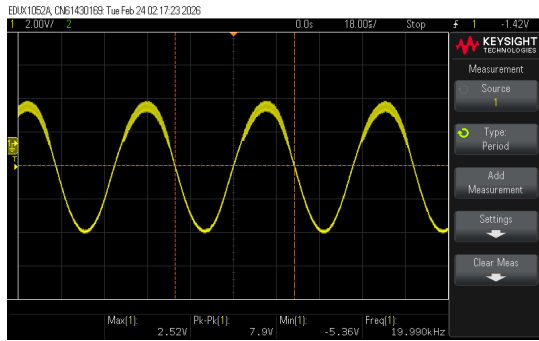


Fig. 27. Hardware output waveform of the power amplifier (Input: 20 mV, Frequency: 20 kHz)

VII. PERFORMANCE ANALYSIS

To evaluate the behaviour of the designed audio amplifier, several important performance parameters were analysed. These parameters help quantify the linearity, noise rejection, and dynamic response of the amplifier. The key metrics considered are Frequency Response, Total Harmonic Distortion (THD), Common Mode Rejection Ratio (CMRR), and Slew Rate.

A. Frequency Response

The frequency response of the complete audio amplifier was analysed using AC sweep simulation in LTSpice. The objective

of this analysis is to verify that the amplifier operates within the required audio frequency band.

The simulation was performed by sweeping the input frequency over a wide range while measuring the magnitude of the output voltage V_{out} . The resulting Bode plot shows the variation of gain with respect to frequency.

From the simulation results, the amplifier maintains a relatively flat gain within the audio frequency band. The gain remains approximately constant between the lower cutoff frequency and upper cutoff frequency, indicating that the amplifier successfully passes signals in the audible range.

The lower cutoff frequency corresponds to the point where the gain decreases by approximately 3 dB from the midband gain. Similarly, the upper cutoff frequency is defined as the frequency where the gain again drops by 3 dB.

The simulated frequency response of the complete amplifier is shown in Figure 28.

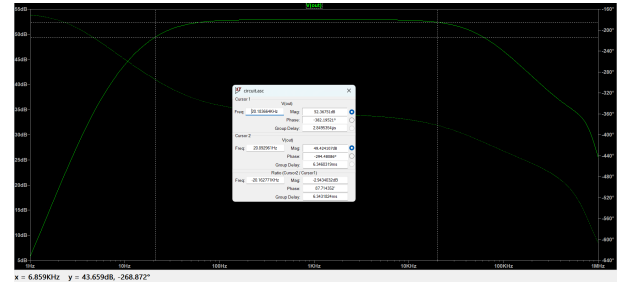


Fig. 28. Frequency response of the complete audio amplifier obtained from LTSpice

B. Hardware Frequency Response Measurement

The frequency response of the complete audio amplifier was also verified experimentally using the Frequency Response Analysis (FRA) feature of the oscilloscope. In this test, a sinusoidal input signal with constant amplitude was applied to the amplifier while the input frequency was swept over a wide range.

The oscilloscope automatically measured the gain and phase shift of the output signal relative to the input signal for different frequencies. This produced the Bode plot showing gain and phase variation with frequency.

From the hardware measurements, the amplifier exhibits a nearly flat gain across the audio frequency band, confirming that the circuit successfully amplifies signals in the intended range. The gain begins to decrease at very low and very high frequencies due to the effects of coupling capacitors and parasitic capacitances in the circuit.

The measured hardware frequency response is shown in Figure 29.

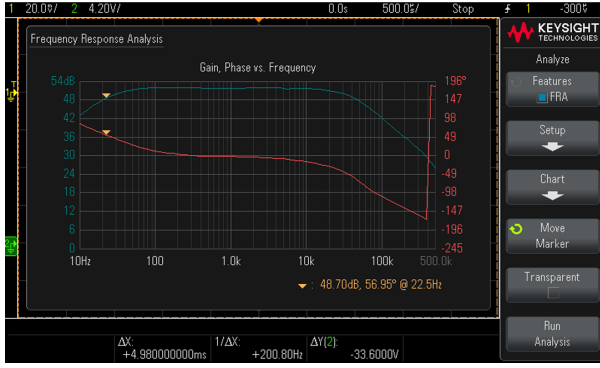


Fig. 29. Hardware frequency response of the complete amplifier measured using the oscilloscope FRA feature

C. Total Harmonic Distortion (THD)

Total Harmonic Distortion (THD) is a measure of the amount of harmonic components introduced by a nonlinear circuit. In an ideal amplifier, the output signal should contain only the fundamental frequency component. However, practical circuits generate additional harmonics which degrade signal fidelity.

THD is defined as the ratio of the RMS value of all harmonic components to the RMS value of the fundamental component:

$$THD = \frac{\sqrt{V_{2,RMS}^2 + V_{3,RMS}^2 + \dots + V_{n,RMS}^2}}{V_{1,RMS}}$$

where

- $V_{1,RMS}$ is the RMS value of the fundamental signal
- $V_{2,RMS}, V_{3,RMS}, \dots$ are the RMS values of the harmonic components

The RMS voltages of the fundamental and harmonic components are obtained from the LTSpice FFT results.

$$V_{1,rms} = 10^{\frac{9.23}{20}} = 10^{0.4615} = 2.894$$

$$V_{2,rms} = 10^{\frac{-17.64}{20}} = 10^{-0.882} = 0.1312$$

$$V_{3,rms} = 10^{\frac{-23}{20}} = 10^{-1.15} = 0.0707$$

$$V_{4,rms} = 10^{\frac{-42.1}{20}} = 10^{-2.105} = 0.0078$$

The total harmonic distortion is defined as

$$THD = \frac{\sqrt{V_{2,rms}^2 + V_{3,rms}^2 + V_{4,rms}^2 + \dots}}{V_{1,rms}}$$

Substituting the obtained values,

$$THD = \frac{\sqrt{(0.1312)^2 + (0.0707)^2 + (0.0078)^2}}{2.894}$$

$$= \frac{\sqrt{0.01721344 + 0.00499 + 0.00006084}}{2.894}$$

$$= \frac{\sqrt{0.02226428}}{2.894}$$

$$THD = 0.0515 \approx 5.15\%$$

The distortion analysis was performed in LTSpice using the Fourier analysis command. The simulation results indicate that the THD of the final output signal is approximately

$$THD \approx 5.16\%$$

This value is within acceptable limits for a discrete audio amplifier and confirms that the Class AB output stage successfully reduces crossover distortion.

D. LTSpice THD Analysis

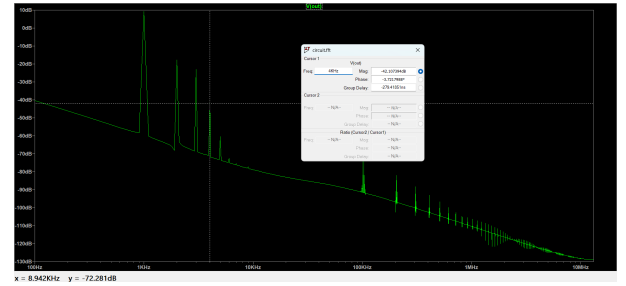


Fig. 30. FFT spectrum of the output signal obtained from LTSpice

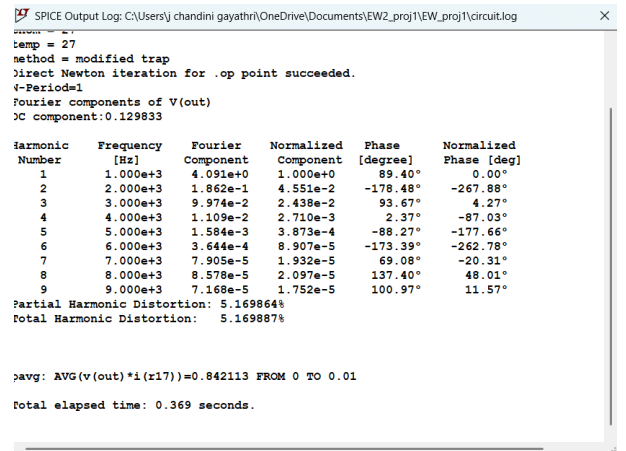


Fig. 31. LTSpice Fourier analysis log showing harmonic components

The total harmonic distortion obtained from LTSpice is

$$THD = 5.16\%$$

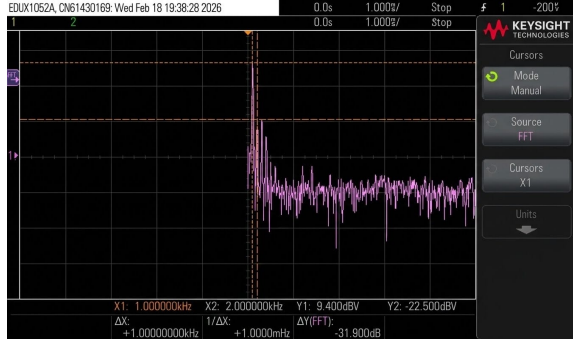


Fig. 32. Hardware simulation of Total Harmonic Distortion (THD)

E. Common Mode Rejection Ratio (CMRR)

Common Mode Rejection Ratio (CMRR) is an important parameter for differential amplifier circuits. It measures the ability of the amplifier to reject signals that appear simultaneously on both inputs while amplifying only the difference between the two input signals.

CMRR is defined as

$$CMRR = 20 \log \left(\frac{A_d}{A_c} \right)$$

where

- A_d is the differential gain
- A_c is the common-mode gain

From the simulation results, the differential gain of the pre-amplifier stage is approximately

$$A_d \approx 20$$

To measure the common-mode gain, the same input signal was applied to both inputs of the differential amplifier. The measured output amplitude was approximately

$$V_{out} = 2.55 \text{ mV}$$

for an input signal of

$$V_{in} = 20 \text{ mV}$$

Thus the common-mode gain becomes

$$A_c = \frac{V_{out}}{V_{in}}$$

$$A_c = \frac{2.55 \text{ mV}}{20 \text{ mV}}$$

$$A_c = 0.1275$$

Substituting these values into the CMRR expression,

$$CMRR = 20 \log \left(\frac{20}{0.1275} \right)$$

$$CMRR = 20 \log(156.86)$$

$$CMRR \approx 43.9 \text{ dB}$$

This value indicates that the differential amplifier effectively suppresses common-mode signals, which helps improve the noise immunity and overall signal quality of the amplifier.

F. Slew Rate (SR)

The slew rate defines the maximum rate at which the output voltage of the amplifier can change with time. It represents how quickly the circuit can respond to rapidly varying input signals.

Mathematically, the slew rate is defined as

$$SR = \max \left(\frac{dV_{out}}{dt} \right)$$

and is typically expressed in volts per microsecond.

The slew rate was obtained from the transient response of the amplifier using LTSpice. From the steepest portion of the output waveform,

$$\Delta V = 959 \text{ mV} = 0.959 \text{ V}$$

$$\Delta t = 37 \mu s$$

Thus,

$$SR = \frac{\Delta V}{\Delta t} = \frac{0.959}{37 \times 10^{-6}} \approx 2.59 \times 10^4 \text{ V/s}$$

$$SR \approx 0.0259 \text{ V}/\mu s$$

or equivalently

$$SR \approx 25.9 \text{ V/ms}$$

This value indicates that the amplifier is capable of responding quickly to changes in the input signal, ensuring faithful reproduction of rapidly varying audio waveforms.

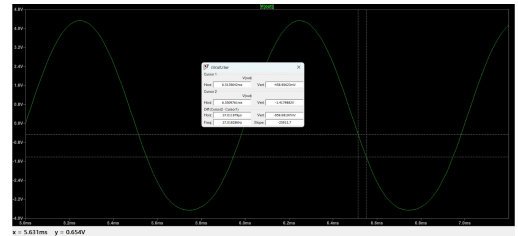


Fig. 33. Slew rate measurement obtained from LTSpice transient analysis

G. Complete LTSpice Circuit Schematic

After designing and verifying each stage individually, all four stages of the amplifier were integrated to form the complete audio amplifier system in LTSpice. The final circuit includes the differential pre-amplifier, common-emitter gain stage, active band-pass filter, and the Class AB power amplifier.

The complete LTSpice schematic of the four-stage audio amplifier is shown in Figure 34.

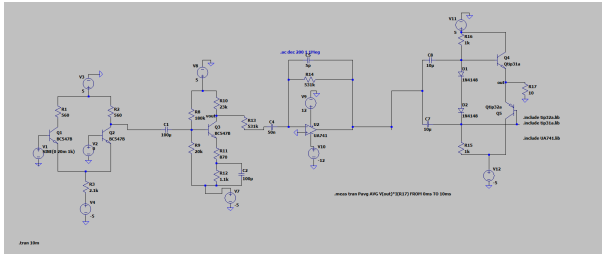


Fig. 34. Complete LTSpice schematic of the four-stage audio amplifier

H. Complete Hardware Implementation

After verifying the operation of each stage through LTSpice simulation, the entire four-stage audio amplifier was implemented on a breadboard using discrete components. The circuit consists of the differential pre-amplifier, common-emitter gain stage, active band-pass filter, and the Class AB power amplifier.

Care was taken to ensure proper grounding, supply connections, and placement of coupling capacitors to avoid unwanted noise and signal distortion. Heat sinks were used for the power transistors in the final stage to safely dissipate power during operation.

The complete hardware implementation of the four-stage audio amplifier is shown in Figure 35.

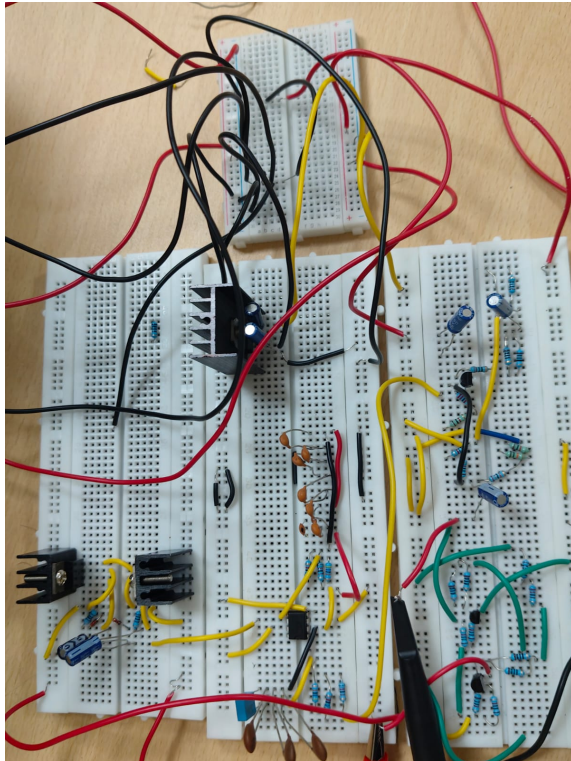


Fig. 35. Complete hardware implementation of the four-stage audio amplifier

I. Discussion

The obtained performance parameters confirm that the designed amplifier operates within the required specifications. The THD remains relatively small, indicating good signal

linearity. The differential pre-amplifier provides adequate rejection of common-mode noise as reflected by the CMRR value. Additionally, the measured slew rate is sufficient for handling the full audio bandwidth.

These results demonstrate that the designed multi-stage amplifier successfully achieves high gain, controlled bandwidth, and reliable output power while maintaining acceptable signal quality.

VIII. CONCLUSION

In this project, a multi-stage audio power amplifier was designed, simulated, and implemented to satisfy the given specifications. The amplifier consists of four major stages: a differential pre-amplifier, a common-emitter gain stage, an active band-pass filter, and a Class AB power amplifier.

The differential pre-amplifier provides the initial amplification of the weak input signal while improving noise immunity. The common-emitter gain stage supplies the major portion of the voltage gain required by the system. An active band-pass filter is used to limit the signal to the audible frequency range, thereby reducing unwanted frequency components. Finally, the Class AB output stage provides the current required to drive a low-resistance load while minimizing crossover distortion.

The circuit was first verified using LTSpice simulations and later implemented using discrete components. The simulation results closely matched the theoretical predictions. The measured output waveforms confirmed that the amplifier is capable of delivering the required output power to a $10\ \Omega$ load while maintaining good signal fidelity.

Overall, the designed amplifier satisfies the major performance requirements, including high voltage gain, adequate bandwidth for audio signals, and stable operation. The project demonstrates the practical design and analysis of a complete analog audio amplifier system using multi-stage amplification techniques.

This project also provided valuable experience in circuit design, simulation, and hardware implementation of analog systems, highlighting the importance of biasing, frequency response control, and power amplification in audio electronics.